Complement-Robust Sensitivity Subspaces for Repeated Quantum Objectives

Molena Huynh[1,*]

[1] *North Carolina State University*

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A sensitivity subspace trained on repeated objectives can fail when a new objective contains components outside the development task span. We introduce a complement-robust second moment that places a development-only innovation radius on the unobserved orthogonal complement. Its minimax discarded-gradient loss has an exact spectral solution. We also prove that $\rho = \operatorname{rank}(\widehat{M})$ transpose-Jacobian products are necessary and sufficient for exact task-only rank-one recovery in a deterministic black-box model, while exact robust completion uses all m observable directions. The evaluation uses depth-one multi-angle MaxCut circuits on ten-vertex neighborhoods drawn from eight checksum-authenticated Stanford Gset graphs. The held-out objective factors have mean development-span distances 0.240 and 0.915 in two protocols. At rank four, complement robustness lowers mean held-out discarded-gradient energy from 0.0590 to 0.0381 for full objectives and from 0.8989 to 0.4396 for localized contrasts. The corresponding unweighted values are 0.4858 and 0.6257. Exact robust action is slower than explicit dense differentiation on every tested instance. The result is an accuracy–cost trade-off for out-of-span transfer, with no runtime advantage at the reported scale.

I. SENSITIVITY LOSS

Let $F : \mathbb{R}^d \to \mathbb{R}^m$ be an observable map at an anchor parameter θ_0, with Jacobian $J = DF(\theta_0)$. A task vector $w \in \mathbb{R}^m$ defines $C_w(\theta) = w^T F(\theta)$ and gradient $J^T w$. For a task second moment $M = \mathbb{E}[ww^T]$ and a rank-r orthogonal projector P, define

$$\mathcal{L}_M(P) = \mathbb{E}\left\|(I-P)J^T w\right\|_2^2 = \operatorname{tr}\left[MJ(I-P)J^T\right]. \quad (1)$$

If $M = LL^T$, the loss is $\left\|(I-P)J^T L\right\|_F^2$. Its minimizer is the dominant rank-r eigenspace of $J^T M J$, with minimum equal to the discarded eigenvalue sum. This is the standard active-subspace construction [1–3].

Development tasks $w_1, \ldots, w_s$ define

$$\widehat{M} = \frac{1}{s}\sum_{i=1}^{s} w_i w_i^T, \qquad \mathcal{S} = \operatorname{range}(\widehat{M}), \qquad \Pi = \Pi_{\mathcal{S}}. \quad (2)$$

For a held-out factor L_t, the dimensionless audit quantity

$$\eta_{\mathcal{S}}(L_t) = \frac{\|(I-\Pi)L_t\|_F}{\|L_t\|_F} \quad (3)$$

separates new task directions from new linear combinations of known directions.

Proposition 1 (span-transfer bound)**.** *For every orthogonal projector P,*

$$\sqrt{\mathcal{L}_{L_t L_t^T}(P)} \le \left\|(I-P)J^T\Pi L_t\right\|_F + \|J\|_2 \|(I-\Pi)L_t\|_F. \quad (4)$$

Proof. Insert $L_t = \Pi L_t + (I-\Pi)L_t$ into the Frobenius representation of Eq. (1), apply the triangle inequality, and use $\|I-P\|_2 = 1$ on the second term. □

* molena.huynh@jmp.com

II. COMPLEMENT-ROBUST CONSTRUCTION

An empirical task moment has no mass on $\mathcal{S}^\perp$. We represent unseen nonnegative mass by the one-sided set

$$\mathcal{U}_\delta = \{\widehat{M} + \Delta : \ 0 \preceq \Delta \preceq \delta(I-\Pi)\}. \quad (5)$$

The radius is fixed without held-out outcomes:

$$\widehat{\delta} = \frac{1}{s}\sum_{i=1}^{s} \|(I-\Pi_{-i})w_i\|_2^2, \quad (6)$$

where Π_{-i} projects onto all development tasks except w_i.

Theorem 2 (exact complement-robust loss)**.** *For every orthogonal projector P,*

$$\sup_{N\in\mathcal{U}_\delta} \mathcal{L}_N(P) = \mathcal{L}_{\widehat{M}}(P) + \delta\,\|(I-\Pi)J(I-P)\|_F^2 = \mathcal{L}_{\widehat{M}+\delta(I-\Pi)}(P). \quad (7)$$

The minimizing rank-r projector is a dominant eigenspace of

$$H_\delta = J^T[\widehat{M} + \delta(I-\Pi)]J. \quad (8)$$

If $s^{-1/2}[w_1, \ldots, w_s] = U\Sigma V^T$ is compact and $U_\perp$ spans $\mathcal{S}^\perp$, then

$$B_\delta = J^T[U\Sigma, \sqrt{\delta}\,U_\perp], \qquad B_\delta B_\delta^T = H_\delta. \quad (9)$$

Proof. Let $R = J(I-P)$ and $K = RR^T \succeq 0$. Feasibility gives

$$\operatorname{tr}(\Delta K) \le \delta \operatorname{tr}[(I-\Pi)K], \quad (10)$$

because $\delta(I-\Pi) - \Delta \succeq 0$. Equality is attained by $\Delta = \delta(I-\Pi)$, proving Eq. (7). The usual trace-maximization argument gives Eq. (8). Finally, $U\Sigma^2 U^T = \widehat{M}$ and $U_\perp U_\perp^T = I - \Pi$, which proves Eq. (9). □

For normalized development tasks, $0 \le \widehat{\delta} \le 1$; it is invariant under task permutation and orthogonal observable-coordinate changes. It is a source-coverage diagnostic, not a distribution-free probability guarantee.

III. DERIVATIVE-QUERY BOUND

Let $\rho = \operatorname{rank}(\widehat{M})$ and define the task-only factor

$$B = J^T U \Sigma \in \mathbb{R}^{d\times\rho}. \tag{11}$$

Each column is one transpose-Jacobian product.

Theorem 3 (exact deterministic query complexity)**.** *In a black-box model returning $J^T v$ for a queried $v \in \mathbb{R}^m$, exactly ρ queries are necessary and sufficient to recover the unique leading task-only sensitivity direction for every J when $d \ge 2$. Exact complement robustness with $\delta > 0$ uses m queries through Eq. (9). Task-only construction costs*

$$O\big(ms\min\{m,s\} + \rho T_{\mathrm{VJP}} + d\rho^2\big) \tag{12}$$

and stores $O(ms + d\rho)$ numbers.

Proof. The identity

$$J^T \widehat{M} J = J^T U \Sigma^2 U^T J = BB^T \tag{13}$$

proves sufficiency. Suppose an adaptive deterministic procedure makes $k < \rho$ queries and receives zero each time. A nonzero $z \in \mathcal{S}$ exists orthogonal to every query. For orthogonal unit vectors $a, b \in \mathbb{R}^d$, the two maps $J_a^T = az^T$ and $J_b^T = bz^T$ produce the same transcript, while their unique leading sensitivity spaces are respectively $\operatorname{span}(a)$ and $\operatorname{span}(b)$. No common output is correct, so $k \ge \rho$. The robust factor has $\rho + (m-\rho) = m$ columns. A compact task SVD, the ρ products, and a thin SVD of the $d \times \rho$ factor give Eq. (12) and its storage count. □

IV. AUTHENTICATED EXPERIMENT

The study uses G1, G6, G11, G14, G18, G22, G32, and G43 from Stanford Gset [4]. Every graph file is accepted only after its SHA-256 digest matches the registered value. A deterministic breadth-first rule selects one ten-vertex induced neighborhood per source. The observable vector contains edge-cut expectations of a depth-one multi-angle QAOA circuit [5–7]. Complex-double state-vector differentiation supplies matched action and dense Jacobians.

Two five-task held-out protocols are fixed before scoring. Full objectives mix every edge observable and have mean span residual 0.240. Localized contrasts emphasize source-absent observable directions and have mean residual 0.915. A rejected vector split, retained only as a leakage control, has maximum residual 6.83×10^{-16}. The rank-four comparison includes task-only, unweighted observable, coordinate-energy, and complement-robust subspaces.

TABLE I. Mean normalized discarded-gradient energy at rank four. Training and held-out columns are reported separately; lower is better.

Protocol	Method	train	held-out
Full objective	Complement-robust	0.0149	0.0381
Full objective	Task-only active subspace	0.0014	0.0590
Full objective	Unweighted observable SVD	0.4658	0.4858
Full objective	Coordinate energy	0.4363	0.4788
Localized contrast	Complement-robust	0.9789	0.4396
Localized contrast	Task-only active subspace	0.0686	0.8989
Localized contrast	Unweighted observable SVD	0.5166	0.6257
Localized contrast	Coordinate energy	0.4864	0.9306

TABLE II. Dimensions and exact transpose-Jacobian counts. Here m is the number of local edge observables, d the parameter dimension, and ρ the development-moment rank.

Graph	m	d	ρ	task VJP	robust VJP
G1	13	23	5	5	13
G6	10	20	5	5	10
G11	12	22	5	5	12
G14	21	31	5	5	21
G18	21	31	5	5	21
G22	9	19	5	5	9
G32	12	22	5	5	12
G43	9	19	5	5	9

Table I exhibits the robustness trade-off. On full objectives, complement completion reduces held-out energy by 35.4% relative to task-only. On localized contrasts, the reduction is 51.1%. Its contrast training loss is much larger, as expected when the model reserves capacity for $\mathcal{S}^\perp$. The method improves the mean on all eight contrast targets relative to task-only; it does not beat every baseline on every graph. Figure 1 displays the corresponding rank path and paired target losses.

The action and dense constructions agree in objective value to at least 1.1×10^{-15} on the rank-four checks. This agreement validates the factorizations in Theorems 2 and 3; it does not establish speed. The task-only method uses five products on every graph, whereas robust completion uses 9–21. On the locked CPU run, robust action is 1.33–1.40 times slower than matched dense robust construction; Fig. 2 and Table II separate derivative counts, wall time, dimensions, and numerical agreement.

V. REPRODUCIBILITY AND SCOPE

The repository records graph addresses and hashes, selected vertex identifiers, task vectors, span residuals, innovation radii, all rank frontiers, adjointness checks, action–dense gaps, derivative counts, and repeated timings. The offline replay recomputes all eight circuits, verifies every registered finding against the locked result, and regenerates the tables and figures.

The evidence concerns ten-qubit, depth-one, noiseless

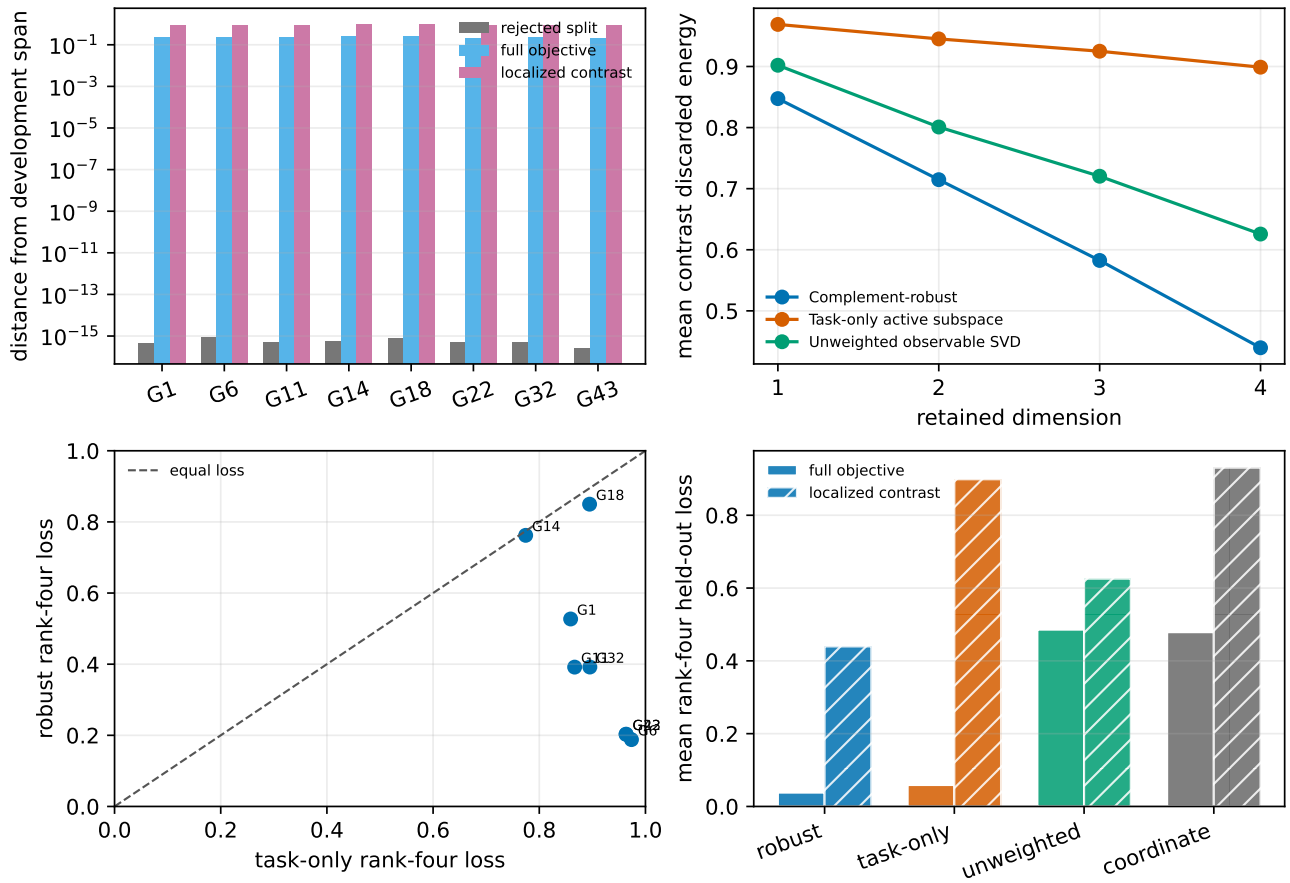


FIG. 1. Out-of-span transfer audit. Top left: development span distance for the leakage control and two registered protocols. Top right: contrast loss across retained ranks. Bottom left: paired rank-four task-only and robust losses; points below the diagonal favor completion. Bottom right: aggregate held-out loss under both protocols.

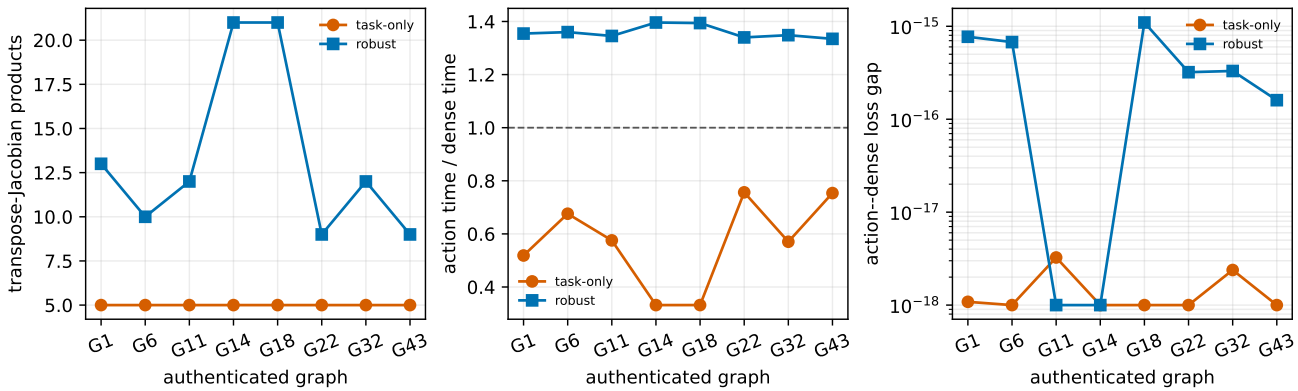


FIG. 2. Derivative and implementation audit. Left: transpose-Jacobian counts predicted by Theorem 3. Center: action-to-dense runtime ratio, with unity marking equal time. Right: action–dense rank-four objective gaps for the factorizations in Theorems 3 and 2.

state-vector objectives. It does not imply hardware performance, global MaxCut quality, or a runtime benefit. Its supported conclusion is narrower: complement-supported moment completion improves out-of-span gradient preservation on the registered graph panel, at the disclosed derivative and wall-clock cost.

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